

Applying the abridged multiplication formulae

The five formulae at work — mental arithmetic, simplification, proofs and identities.

LESSON 108–109

2 × 40 MIN

MATEMATIKA 7, §49

S8 2.4

LESSON CLOCK

10'

20'

25'

25'

5'

The five formulae in one table	10 min
Mental arithmetic	20 min
Simplifying	25 min
Proving something	25 min
Homework	5 min

LEARNING OBJECTIVES

- Choose the right formula from the shape of an expression.
- Use the formulae to calculate quickly without a calculator.
- Simplify expressions in which several formulae appear together.
- Prove a simple divisibility statement by factorising.

TEXTBOOK

National	pp. 319–326
Cambridge	Stage 8, pp. 25–30
Workbook	Exercise 2.4

01

Explanation

The five formulae in one table

NAME	FORMULA	RECOGNISE IT BY
square of a sum	$(a + b)^2 = a^2 + 2ab + b^2$	three terms, middle $2ab$
square of a difference	$(a - b)^2 = a^2 - 2ab + b^2$	three terms, middle $-2ab$
difference of squares	$a^2 - b^2 = (a - b)(a + b)$	two square terms, a minus
cube of a sum or difference	$(a \pm b)^3$	four terms, $1, 3, 3, 1$
sum or difference of cubes	$a^3 \pm b^3$	two cube terms

COUNT THE TERMS FIRST

Two terms point to a difference of squares or a sum of cubes; three to a square; four to a cube or to grouping. Counting decides the formula before any algebra is attempted.

Mental arithmetic

TASK	FORMULA	WORKING	ANSWER
103^2	$(a + b)^2$	$10000 + 600 + 9$	10609
97^2	$(a - b)^2$	$10000 - 600 + 9$	9409
$63 \cdot 57$	$a^2 - b^2$	$3600 - 9$	3591
$75^2 - 25^2$	$a^2 - b^2$	$50 \cdot 100$	5000
31^3	$(a + b)^3$	$27000 + 2700 + 90 + 1$	29791

75² – 25² IN ONE LINE

Factorised it is $(75 - 25)(75 + 25) = 50 \cdot 100$. Squaring both numbers first takes four times as long and invites an arithmetic slip.

Simplifying

EXPRESSION	SIMPLIFIED
$(a + b)^2 - (a - b)^2$	$4ab$
$(a + b)^2 + (a - b)^2$	$2a^2 + 2b^2$
$(x + 3)^2 - (x - 3)^2$	$12x$
$(x + 2)(x - 2) - x^2$	-4
$\frac{x^2 - 16}{x + 4}$	$x - 4$

EXPAND BOTH BRACKETS BEFORE SUBTRACTING

$(a + b)^2 - (a - b)^2$ is not $(2b)^2$. Subtracting the brackets as if they were single numbers is the standard slip; expand each one and then subtract term by term.

Proving something

A factorised form can prove a statement that the expanded form hides completely.

CLAIM	FACTORISED	WHY IT FOLLOWS
$n^2 + n$	$n(n + 1)$	consecutive numbers — one is even
$n^2 - 1$	$(n - 1)(n + 1)$	for odd n , both factors are even
$(n + 1)^2 - n^2$	$2n + 1$	always odd
$n^3 - n$	$(n - 1)n(n + 1)$	three consecutive numbers

WHY THE DIFFERENCE OF TWO CONSECUTIVE SQUARES IS ODD

$(n + 1)^2 - n^2 = 2n + 1$, which is odd for every whole n . It also explains why every odd number is a difference of two squares — $9 = 5^2 - 4^2$, $11 = 6^2 - 5^2$, and so on.

02 Worked examples

EXAMPLE 1 Find $75^2 - 25^2$ without a calculator.

1 A difference of squares.

2 $= (75 - 25)(75 + 25)$

3 $= 50 \cdot 100$

4 $= 5000$

ANSWER

5000

EXAMPLE 2 Simplify $(x + 3)^2 - (x - 3)^2$.

① $(x + 3)^2 = x^2 + 6x + 9$

② $(x - 3)^2 = x^2 - 6x + 9$

③ Subtract: $12x$.

The squares and the 9s cancel.

ANSWER

$12x$

EXAMPLE 3 Prove that $n^2 - 1$ is divisible by 8 for every odd n .

① $n^2 - 1 = (n - 1)(n + 1)$

② For odd n both factors are even.

③ They are consecutive even numbers, so one is a multiple of 4.

④ The product is therefore a multiple of $2 \cdot 4 = 8$.

ANSWER

Divisible by 8

03 Interactive model

Give the class $25 \cdot 35$, $51 \cdot 49$ and 98^2 as a race against the calculators; the formulae win, and the point needs no further argument.

Which formula fits? $x^2 - 49$ fits: $x^2 + 10x + 25$ fits: $x^3 - 27$ fits:

$x^3 - 6x^2 + 12x - 8$ fits:

difference of cubes

cube of a difference

square of a difference

none

97^2 is easiest as:

$(100 - 3)^2$

$(90 + 7)^2$

$97 \cdot 97$

$(100 + 3)^2$

$63 \cdot 57$ equals:

3591

3600

3609

3549

$(a + b)^2 - (a - b)^2$ equals:

$2b^2$

$4ab$

$2a^2$

0

$(n + 1)^2 - n^2$ is:

always even

always odd

always prime

sometimes zero

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Abridged multiplication	Qisqa ko'paytirish	Сокращённое умножение
To apply	Qo'llash	Применять
To simplify	Soddalashtirish	Упростить
Identity	Ayniyat	Тождество
Divisible	Bo'linadi	Делится
Consecutive	Ketma-ket	Последовательные
Even number	Juft son	Чётное число
Proof	Isbot	Доказательство

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Three terms with a middle of $2ab$ point to:

a square

a difference of squares

a cube

grouping

Two square terms with a minus point to:

a square of a sum

a difference of squares

a sum of cubes

nothing

$75^2 - 25^2$ equals:

2500

5000

5625

10000

$(x + 3)^2 - (x - 3)^2$ equals:

$12x$

18

$6x$

0

$n^2 + n$ is always:

odd

even

prime

square

103^2 equals:

10609

10309

10906

10009

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 103^2

ANSWER 10609

2. 97^2

ANSWER 9409

3. $63 \cdot 57$

ANSWER 3591

4. $75^2 - 25^2$

ANSWER 5000

5. $(x + 3)^2 - (x - 3)^2$

ANSWER $12x$

6. $(x + 2)(x - 2) - x^2$

ANSWER -4

7.
$$\frac{x^2 - 16}{x + 4}$$

ANSWER $x - 4$ **Medium · the standard question**

7 PROBLEMS

1. $(a + b)^2 - (a - b)^2$

ANSWER $4ab$

2. $(a + b)^2 + (a - b)^2$

ANSWER $2a^2 + 2b^2$

3. 31^3

ANSWER 29791

4. $45^2 - 35^2$

ANSWER 800

5. $(2x + 1)^2 - (2x - 1)^2$

ANSWER $8x$

6. Prove $n^2 + n$ is even

ANSWER $n(n + 1)$ — one factor is even

7. $(n + 1)^2 - n^2$

ANSWER $2n + 1$

Hard · stretch and reason

7 PROBLEMS

1. Prove $n^2 - 1$ is divisible by 8 for odd n

ANSWER Two consecutive even factors, one a multiple of 4

2. $n^3 - n$ factorised

ANSWER $(n - 1)n(n + 1)$

3. Write 11 as a difference of two squares

ANSWER $6^2 - 5^2$

4. $(a + b)^3 - (a - b)^3$

ANSWER $6a^2b + 2b^3$

5. If $a + b = 10$ and $a - b = 4$, find $a^2 - b^2$

ANSWER 40

6. $1000^2 - 999^2$

ANSWER 1999

7. Simplify $\frac{x^3 - 27}{x - 3}$

ANSWER $x^2 + 3x + 9$

07 Homework

Homework — 5 tasks

Count the terms, name the formula, then calculate.

- Find 104^2 , 96^2 and $72 \cdot 68$ by the formulae.
- Find $85^2 - 15^2$ in one line.
- Simplify $(x + 5)^2 - (x - 5)^2$ and $(3a + 1)^2 - (3a - 1)^2$.
- Prove that $n^3 - n$ is divisible by 6 for every whole n .

5. Write 15 and 21 each as a difference of two squares.